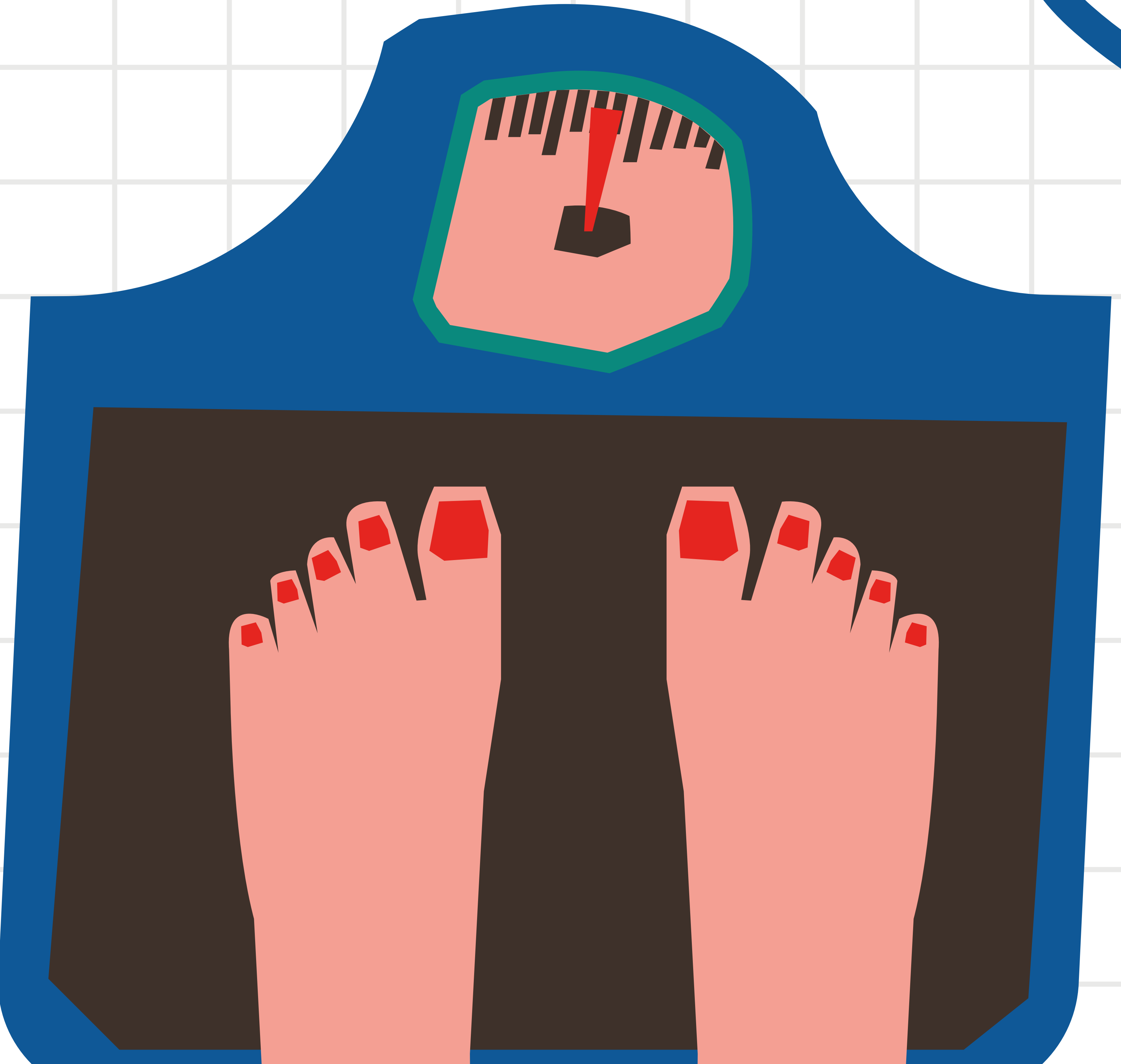


# PREDICTING OBESITY STATUS

By: Kiara Cueva, Caleb Tran, Cynthia Du, and Sonia Kang

# LEARNING CONTENT

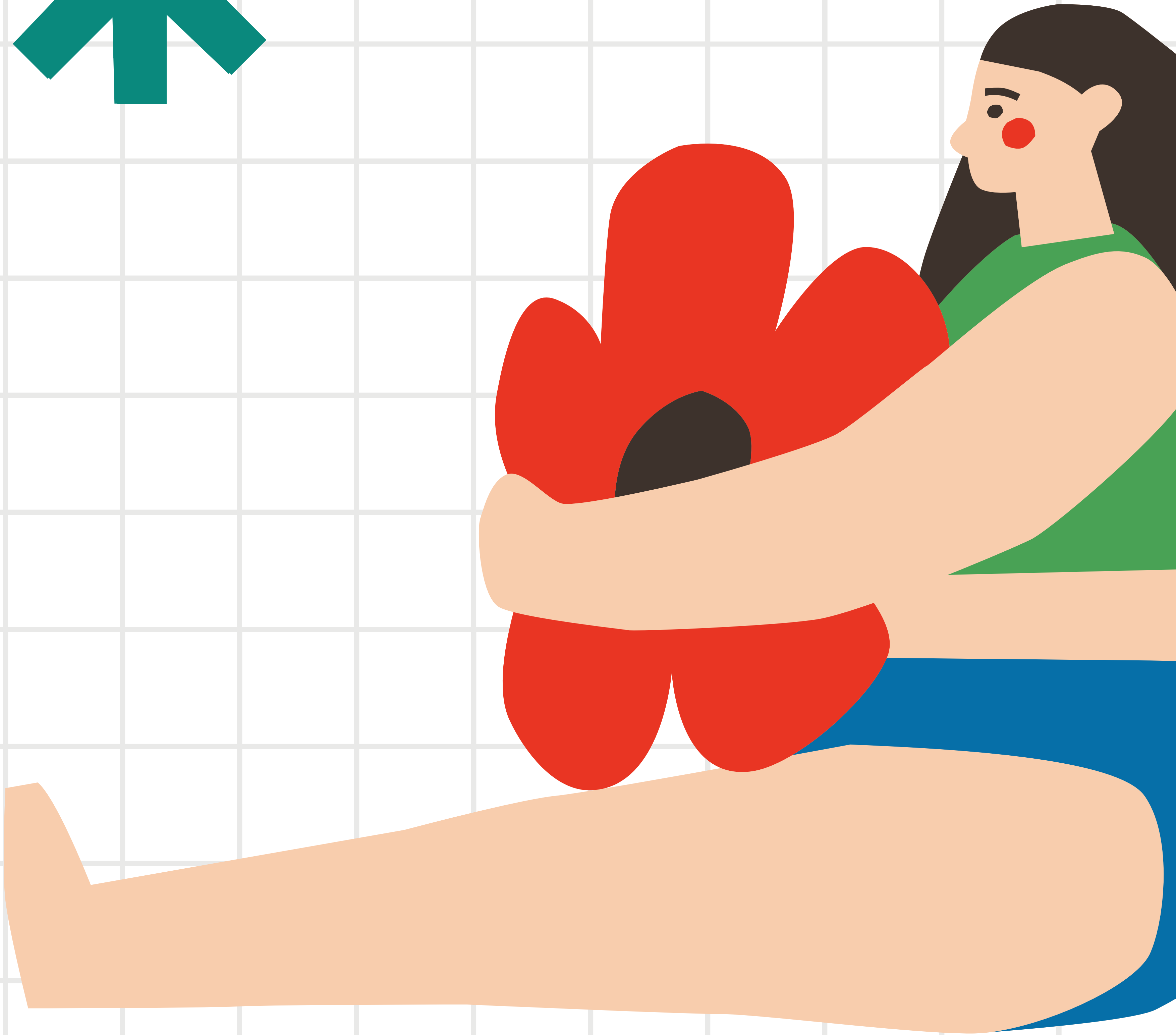
1. Introduction
2. Methodology
3. Results & Discussion
4. Limitations & Conclusions

# INTRODUCTION

Context of our Study and of the Dataset

# INTRODUCTION

Obesity is a chronic disease. The term ‘obese’ describes somebody who has excessive fat deposits in their body. Predicting obesity is important because it allows for early intervention and targeted strategies to mitigate associated health risks, such as cardiovascular diseases, diabetes, and certain types of cancer.



# OBESITY DATASET

## 32014 observations

Each observation represents an individual subject with associated features and an outcome variable of their obese status.

## 30 variables

Detailed information recorded with each subject (e.g. cholesterol level, resting blood pressure, race)

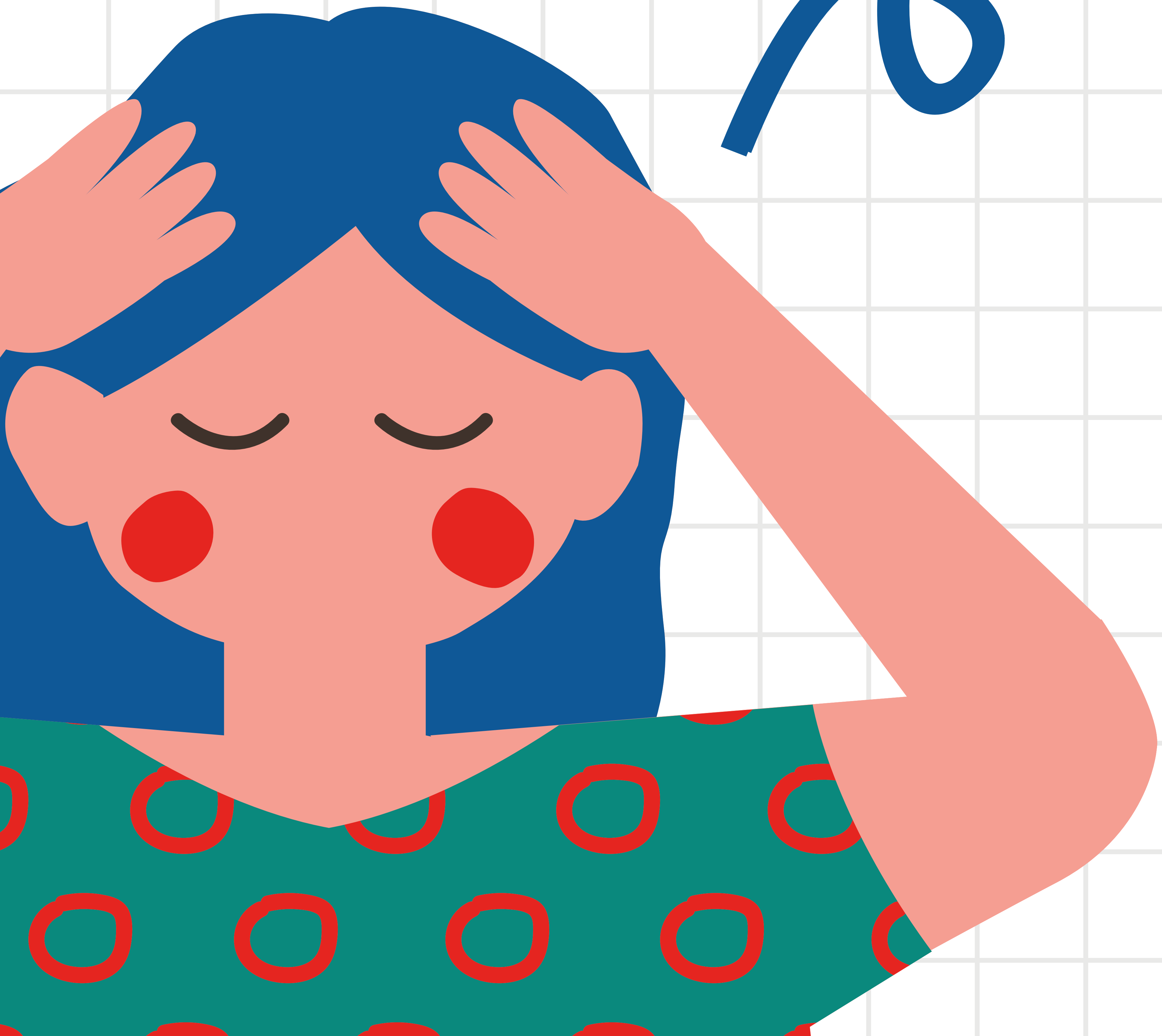
# METHODOLOGY

Data Cleaning and Modeling Process

# THE PROCESS

| 1.Clean Data | 2. Create and Analyze Models | 3. Compare Models |
| --- | --- | --- |
| Dealing with NA's and multicollinearity | Undergo feature selection and find misclassification rates | Choose model with best misclassification rate and simplicity |

# CLEANING DATA



**Dealing with NA's**

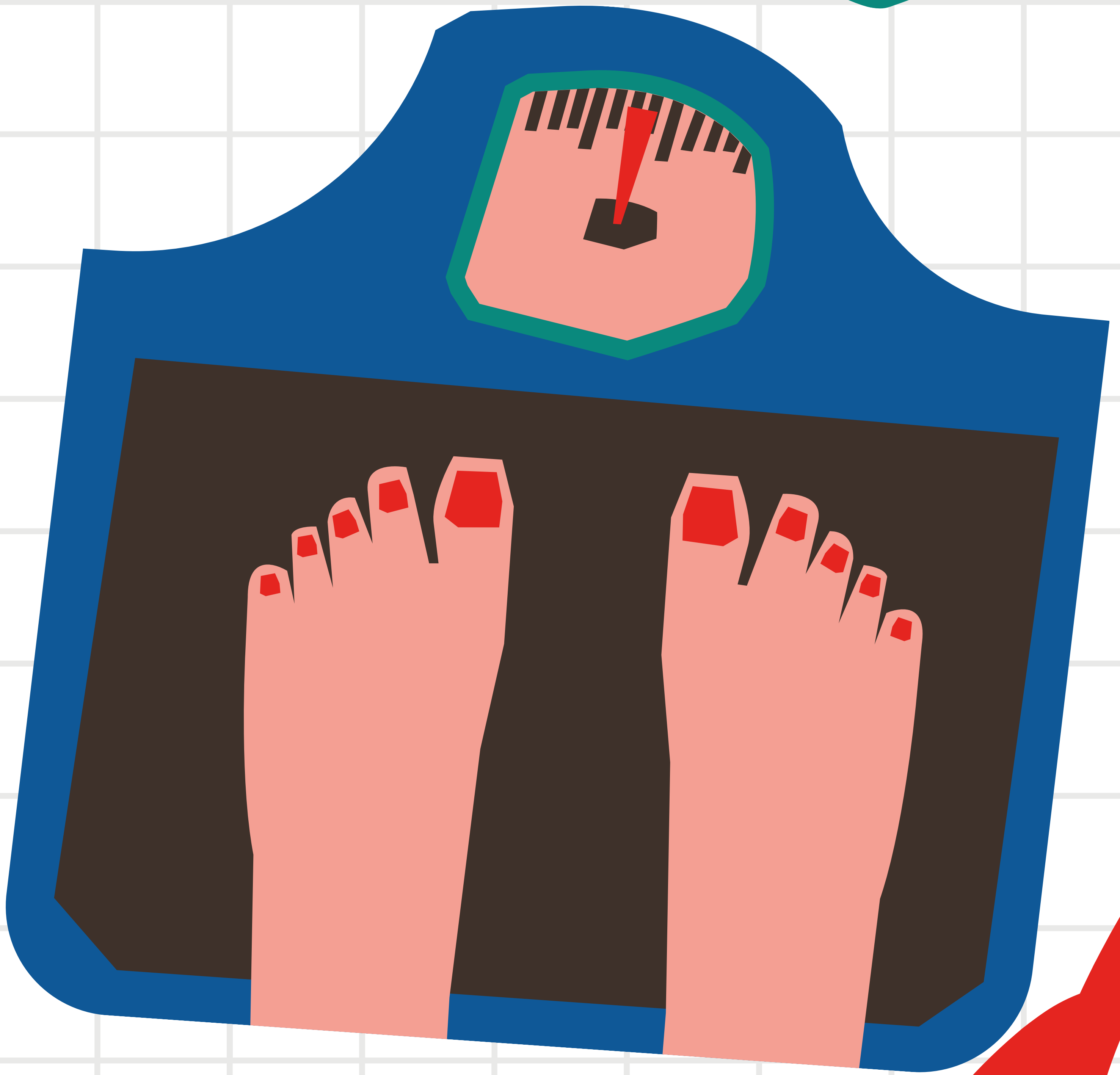
- For all numerical columns, replace NA's with the **median** of that column
- For all categorical columns, replace NA's with the **mode** of that column

This imputation was done on both training and testing data sets

# CLEANING DATA

All predictors had a small proportion of missing values. Thus, we did not need to eliminate any predictors immediately and could impute the missing values as stated previously.

The percentage of missing values in each column ranged from 7.496% to 8.602%. Imputing the median and mode respectively would be representative of the overall distribution of each column's data



# VARIABLE SELECTION

## AIC

Balances model complexity and fit, preferring models that explain the data well without being overly complex

## BIC

BIC is similar to AIC but applies a stronger penalty for model complexity. Good when working with larger datasets or when trying to avoid overfitting

# VARIABLE SELECTION CONT.

To evaluate the importance of each variable we looped through the variables and fit a logistic regression model to each individual variable to predict Obesity Status. We then extracted the AIC and BIC metrics. The top ten most relevant variables are listed below:

| Variable | AIC | BIC |
|---|---|---|
| family_history_with_overweight | 40848.65 | 40865.40 |
| FAF | 40975.70 | 40992.44 |
| CAEC | 41330.75 | 41364.25 |
| CH20 | 41521.49 | 41538.23 |
| FAVC | 41560.79 | 41577.54 |
| MTRANS | 41875.93 | 41917.79 |
| FCVC | 42046.29 | 42063.04 |
| Gender | 42289.83 | 42306.58 |
| SCC | 43241.27 | 42358.02 |
| CALC | 42436.82 | 42470.31 |

# CHOOSING A MODEL

We tested the following models:

## Logistic Regression

### PROS

- simple and interpretable
- no distributional assumptions

### CONS

- not very flexible
- can be inaccurate if boundary isn't linear

## LDA

### PROS

- dimensionality reduction
- accounts for feature importance

### CONS

- assumes Gaussian distribution and equal covariance among classes

## Random Forest

### PROS

- very flexible, handles complex relationships

### CONS

- complicated and less interpretable
- computationally expensive

# LOGISTIC REGRESSION MODEL

We constructed a logistic regression model on predicting obesity status using 22 predictors we selected through forward stepwise variable selection based on AIC.

## CONFUSION MATRIX

| PREDICTED | ACTUAL: NOT OBESE | ACTUAL: OBESE |
|---|---|---|
| NOT OBESE | 1511 | 443 |
| OBESE | 265 | 714 |

ACCURACY: 75.86%

MCR: 24.14%

# LDA MODEL WITH CV 10-FOLD

We constructed an LDA Model with the CV 10-Fold Model to predict obesity status with the following 6 selected variables:

"Cholesterol", "CH2O", "FCVC", "family_history_with_overweight", "CAEC", "FAVC"

## CLASSIFICATION MATRIX

| | Not Obese | Obese |
|---|---|---|
| Not Obese | 9596 | 2959 |
| Obese | 2325 | 4597 |

**ACCURACY: 72.797%**

**SENSITIVITY: 0.8050**

**LDA ERROR RATE : 0.271**

**SPECIFICITY: 0.6084**

# RANDOM FOREST MODEL

Our final attempt was the Random Forest Model.

After imputing missing values, we optimized the Random Forest model by tuning the number of trees and mtry values and analyzed variable importance to balance complexity and performance.

## CLASSIFICATION MATRIX

| | Not Obese | Obese |
|---|---|---|
| Not Obese | 15615 | 10 |
| Obese | 104 | 9883 |

## ACCURACY: 99.78

## MISCLASSIFICATION RATE : 0.0081

# REFINING THE RF MODEL

1

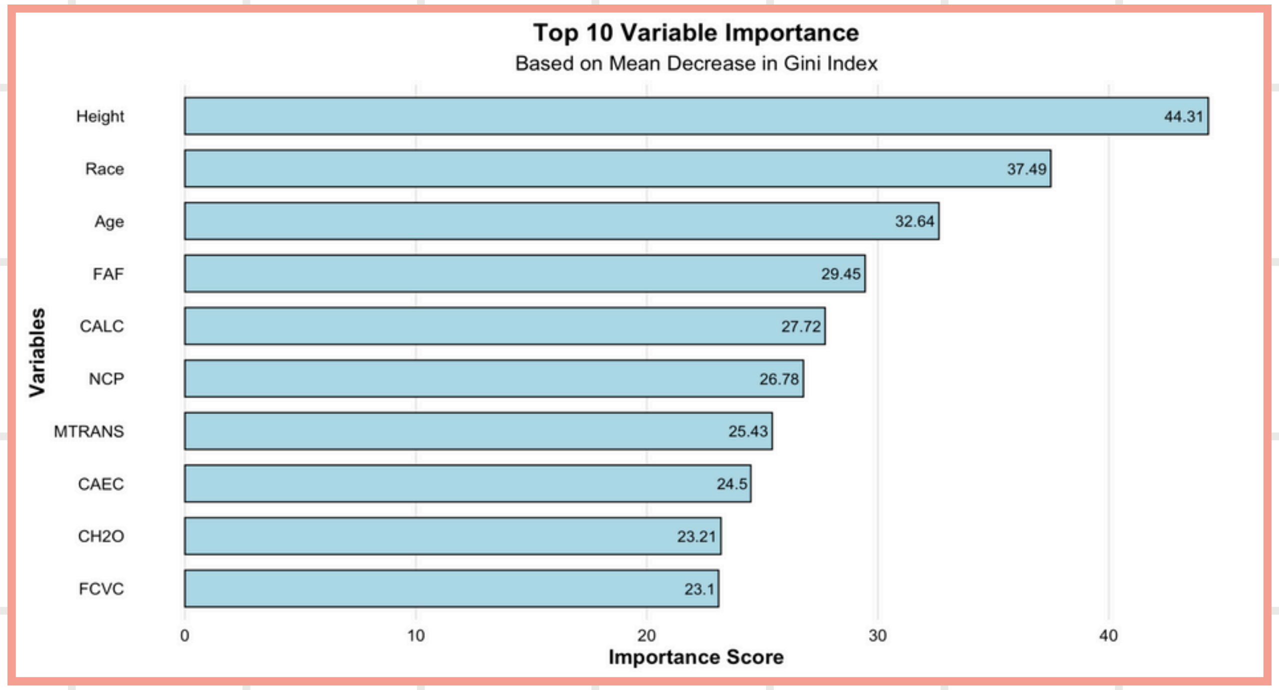
Variable Importance :
Found top ten variables Using plots and indexing

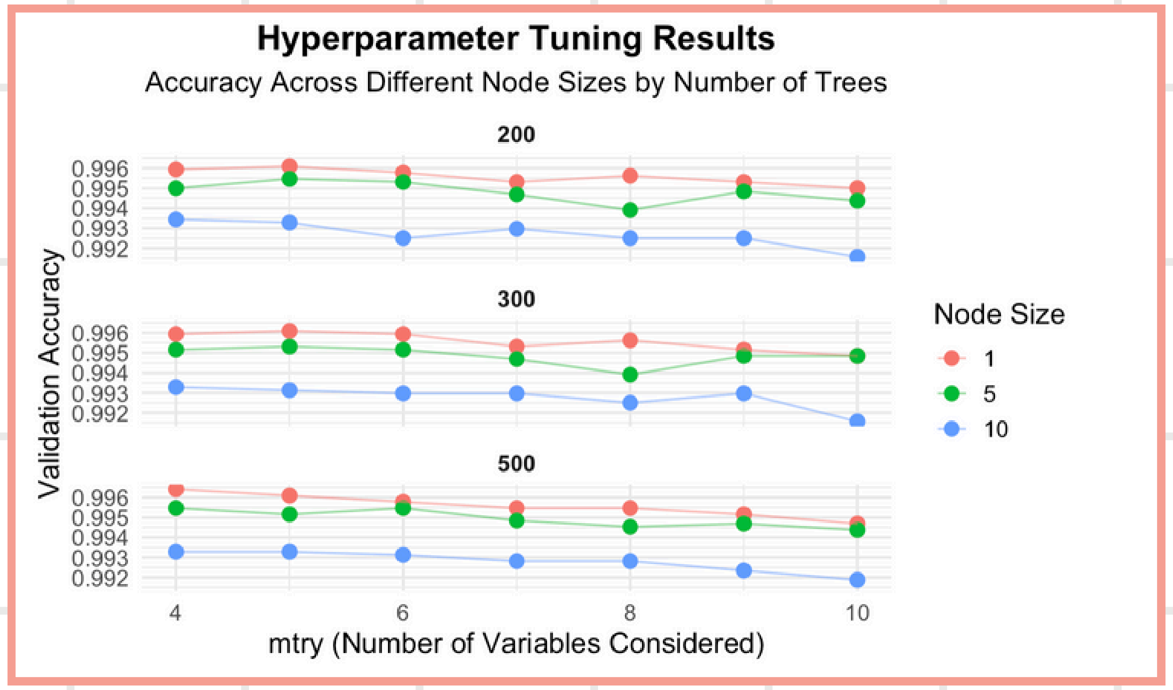
2

Mtry, ntree, nodesize:
Filtered through different combinations of model and filtered the best one

3

validation and error comparison:
Looked at model metrics and tried to reduce complexity



# RESULTS & DISCUSSION

Analysis of Our Final Model

# OUR MODEL AT A GLANCE

| | | |
|---|---|---|
| **Model:** Random Forest | **Number of Predictors Used:** 10 | **Kaggle Rank:** 15 |
| **Accuracy:** 0.997 | **MCR:** 0.0081 | **R^2:** 0.9263 |

# LIMITATIONS & CONCLUSIONS

Assumptions, Next Steps, and Closing Remarks

# LIMITATIONS

## 1. Data Cleaning

- imputed missing values with median but data may not be Normally Distributed
- may not have chosen ideal variables
- potential collinearity

## 2. Model Building

- may have chosen the wrong mtry
  - will lead to overfitting
- didn't scale features, model may be too sensitive as a result

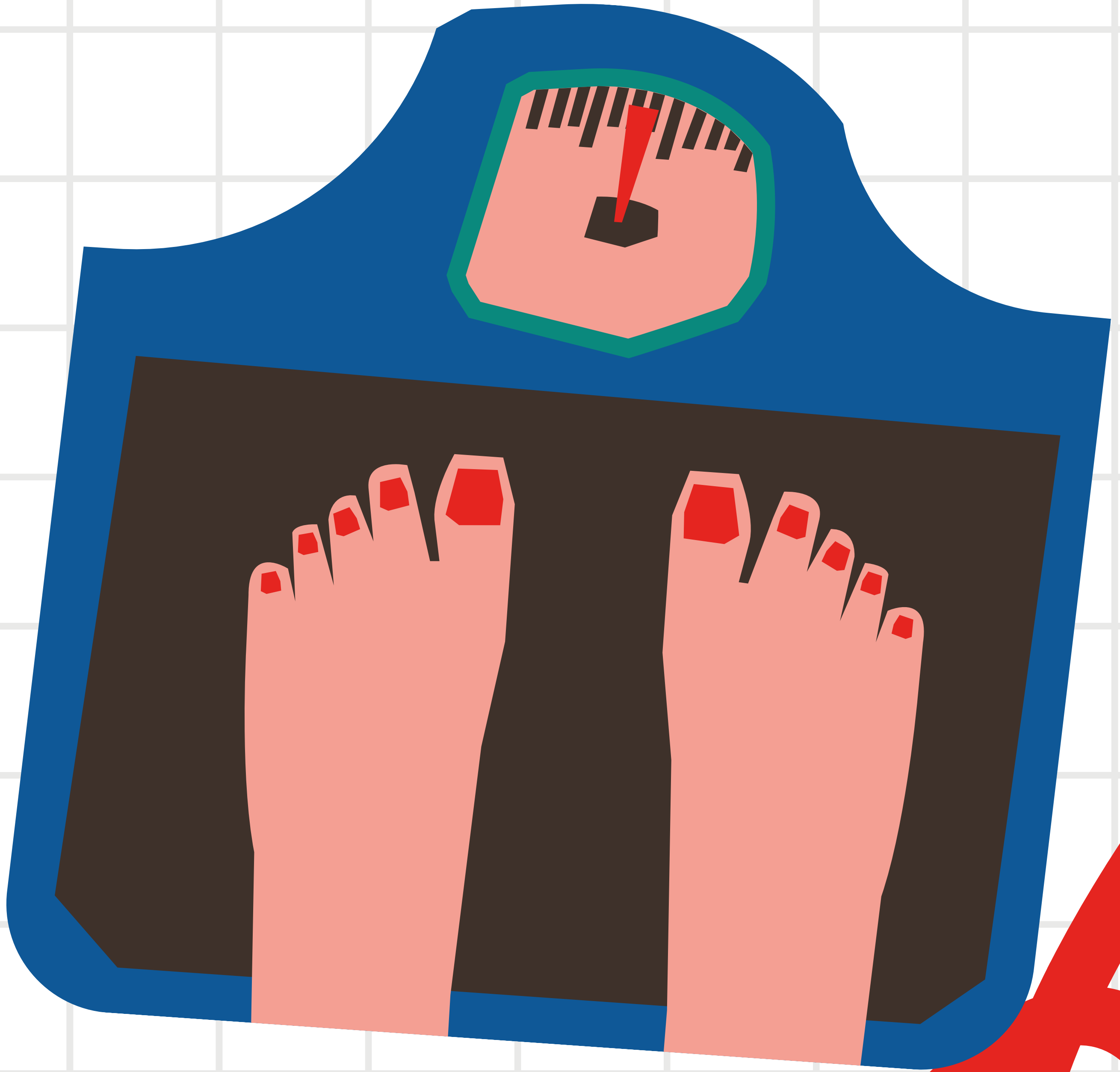
## 3. Assumptions

- assumed independence between observations
- assumed no multicollinearity (after we checked)
- assumed relevance and additivity of features

# CONCLUSION

Overall, using our model, we get a very accurate prediction of obesity status. However, we acknowledge that our model may be limited by imputation in data cleaning and our assumptions about how the observations were gathered and the relationship between the predictors.

In our next steps we would conduct further research on the study and on the variables that could deepen our understanding of the issue and help us fine-tune our model.



# THANKS FOR LISTENING